SkillNavora · Product Management Master Handbook

Module 7: Product Analytics and Metrics

Detailed supplement: Goodhart's Law, the 2×2 metric architecture, guardrails, growth loops, retention curves and the OKR cascade

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Official site: www.skillnavora.in

Prepared for Skillnavora — Learn. Practise. Implement. Grow.

MODULE 7 · HOW TO USE THIS SUPPLEMENT

Coverage map: what is new and what is already in the module

Read this supplement next to the main Module 7 handbook and the detailed PDF. **Extended** means the handbook has the topic and this PDF adds the deck's example. **New** means the handbook did not have it. Company examples come from the deck and are not verified; check them before you quote them.

Topic	Already in the module?	Status here
Ladder of metrics: input and output	Yes (ladder section)	Extended: the causal link and what you can control
Leading and lagging metrics	Yes	Extended: examples side by side
North Star metric	Yes	Extended: the 'one action that creates value' test
Goodhart's Law and metric gaming	Partly (one mention)	New: proxy displacement and three PM traps
2×2: controllability and timing	No	New: the four quadrants
Counter-metric and guardrail architecture	Yes	Extended: a table of focus areas, primary metrics and guardrails
AARRR and HEART	Yes	Not repeated here
Growth loops versus linear funnels	No	New: viral, content and other loop types
Retention curve shapes	Partly	New: four shapes and how to read them
Google Maps metrics architecture (activity)	No	New: a worked model for two stakeholders
OKR cascade from North Star to sprints	Partly (OKR section)	Extended: four levels with one worked chain

NOTE ON THE SOURCE

The deck's Goodhart's Law and guardrail slides include numeric targets such as '<2 percent' or '>=25 percent'. They are illustrative. Set your own targets from your own baseline.

NEW · METRIC GAMING

Goodhart's Law: when a measure becomes a target

The deck's version of the idea: **'When a measure becomes a target, it ceases to be a good measure.'** Three mechanisms cause the damage:

Mechanism	What happens
Proxy displacement	Teams confuse a proxy (clicks, raw daily active users) with real customer value (completing the core task).
Path of least resistance	Under sprint or OKR pressure, teams take the easiest way to move the number.
Compounding side-effects	Unmonitored dimensions decline quietly until trust, retention or unit economics fall.

Three PM traps (illustrative)

Trap	Example from the deck	Lesson
Notification trap	Five times more promotional push alerts raise daily active users by 8 percent, while uninstalls and muted alerts rise by 40 percent.	Watch the guardrail: uninstalls and mutes.
Signup friction trap	Pre-ticking a newsletter box lifts registrations, and then the new users unsubscribe.	Count the quality of signups, not only the number.
Click-farm trap	Acquisition targets reached with low-intent or bot traffic.	Pair acquisition with retention and fraud guardrails.

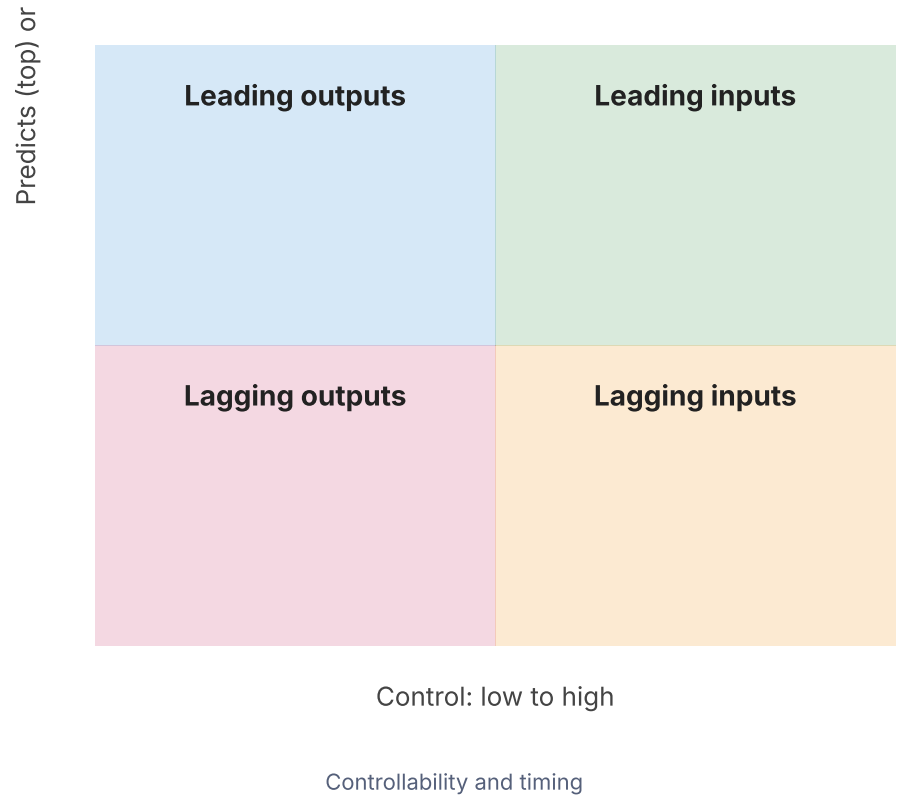
NEW · METRIC ARCHITECTURE

The 2×2: controllability versus timing

Sort every metric on two questions: can your team **control** it, and does it **predict** the future or **report** the past? That gives four quadrants.

Quadrant	What it means	Example from the deck (illustrative)
Leading inputs (high control, predictive)	The daily levers your team controls	Share of documents auto-verified on first submission (fintech); autocomplete latency (e-commerce)
Leading outputs (low control, predictive)	Early behaviour that signals long-term value	A team reaching a 'magic moment' in its first weeks; a user doing a key action on two devices
Lagging inputs	Activity you can control, but its results take time to show	Changes to onboarding flows measured on later cohorts
Lagging outputs (low control, reports the past)	Final results	Revenue; retained users at 90 days

Start with the leading inputs. They are what the team can change this week.



Controllability and timing

EXTENDED · GUARDRAILS

Primary metric, gaming risk and counter-guardrail

Every primary metric needs a guardrail: the number you refuse to damage while you push the first one. The deck's pattern:

Focus area	Primary metric (what we push)	Gaming risk (what can break)	Counter-guardrail (what we defend)
Acquisition	New signups and app installs	Low-intent or bot signups	Day-30 retention; fraud and chargeback rate
Engagement	Daily active users	Over-notifying users to create artificial activity	App uninstalls per 1,000 sessions; notification mute rate
Monetisation	Average revenue per user	Dark patterns and hidden renewals	Support ticket ratio; refund and dispute rate

Targets in the table are examples. Set your own from your baseline.

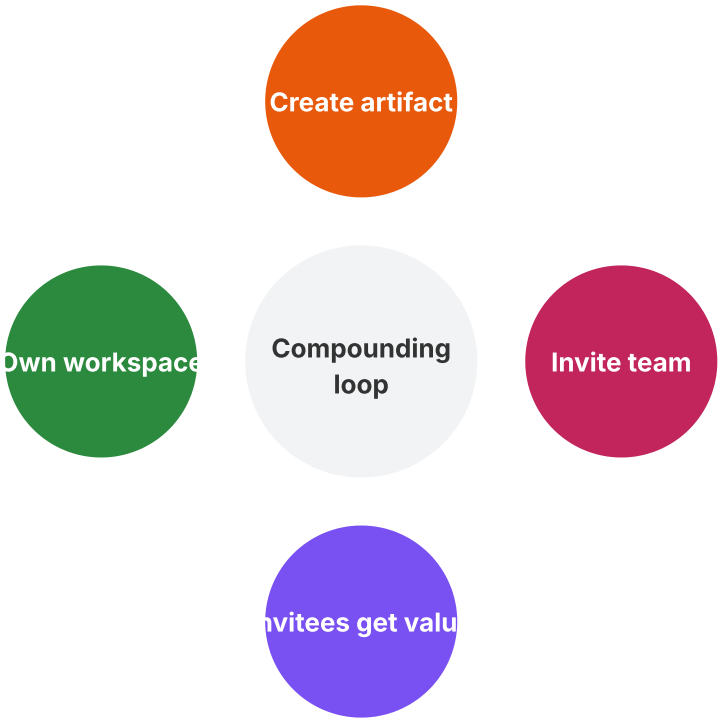
NEW · GROWTH

Linear funnels versus compounding loops

A linear funnel (acquisition, activation, retention, revenue) needs new input at every step. A loop feeds its own output back in: the actions of current users bring in and activate the next group.

Loop type	How it works (deck)	Metric to track
Viral or collaboration loop	A user creates an artifact, invites team members, and they start their own workspaces.	Viral coefficient above 1.0; loop cycle time
Content or search loop	An active user posts a question or review; the platform publishes a crawlable page; search engines index it; new visitors arrive and some join.	Pages indexed; share of new users from search

The deck names collaboration products such as Figma, Slack, Miro and Notion as examples of the viral loop. Check how each one works today before you use it as a case study.



A viral loop, from the deck

NEW · RETENTION CURVES

Four retention shapes and what they mean

Shape	What it looks like	What it tells you
Flat	Almost no drop-off; 95 of 100 users remain at month 12.	Strong habit or a core need; check the sample first.
Cliff then tail	Loses most users early, then the few who stay remain for years.	Fix onboarding and the first value moment.
Dip then recover	Flat for a few months, drops, then levels off again.	Find out what brought users back.
Steady decline	A constant downward line that never flattens; users are almost gone by month three.	No retained core; rethink the problem before growth spend.

Practice: for a product you use, draw the shape you expect, then check the real curve in analytics. Your guess is data about your own assumptions.

NEW · ACTIVITY

Google Maps metrics architecture

The deck's activity asks for metrics for two groups of stakeholders. Here is the model answer for drivers and commuters (illustrative):

Element	Answer
Core job to be done	Find the fastest, safest route with a reliable arrival time.
North Star metric	Share of trips taken on the best route (time, road condition, tolls, live traffic).
Leading metric	Number of navigation sessions started.
Lagging metric	Weekly active drivers; 90-day retention.
Input	Reroute time under two seconds; speed of hazard reports.
Output	Share of trips completed on the best route; trip satisfaction.
Counter-guardrails	Trip abandonment under 2 percent; arrival-time error under 3 minutes; distracted-driving alerts.

Then do the second group yourself: local businesses. Their job is to be found nearby and turn visits into paying customers. Write its North Star and guardrails.

EXTENDED · OKR CASCADE

From North Star to sprint

The deck shows four levels. Each level narrows the one above it. The example uses a music streaming company (the deck's figures are illustrative):

Level	What it is	Example from the deck
1. Company North Star	A 1 to 3 year metric	Total monthly listening hours
2. Input drivers	Levers assigned to teams for the year	Discovery breadth; personalisation depth; sharing
3. Team OKRs	Quarterly objective with measurable key results	Search team: make every search deliver a playable track; cut time to first song
4. Initiatives and sprints	The weekly work that moves the key results	Ship a faster search results screen

Key results must be measurable, time-bound and specific. 'Make search faster' is not a key result. 'Reduce time to first song from 4.2 seconds to 1.8 seconds this quarter' is.



From North Star to sprint

ASSIGNMENT

Build a metric architecture for a product

1. Pick a product you use. Name its users and their jobs to be done.
2. Write one North Star metric and explain why it captures the job.
3. Build a metric ladder with at least two leading metrics, two lagging metrics, and one input-to-output link.
4. Add a counter-guardrail for each primary metric, and say how it could be gamed.
5. Repeat the exercise for a second stakeholder group.

SELF-CHECK

Is each metric linked to a job? Did you name one way it could be gamed? Are your targets from a baseline, not invented?

Quiz with answers

1. State Goodhart's Law in the deck's words.
   Answer: 'When a measure becomes a target, it ceases to be a good measure.'
2. What are the two questions of the 2×2?
   Answer: can the team control the metric, and does it predict the future or report the past?
3. What is a counter-guardrail?
   Answer: a metric you refuse to damage while you push the primary metric.
4. How does a loop differ from a linear funnel?
   Answer: a loop feeds its output back in to bring in and activate the next group.
5. Which retention shape suggests a strong habit?
   Answer: the flat curve, after checking the sample.
6. Which level of the OKR cascade is a quarterly objective?
   Answer: level 3, team-level OKRs.

MODULE 7 · FINAL

Completion checklist

Completion checklist

Done	Item
☐	Read the coverage map and marked the new topics
☐	Named one metric-gaming risk for a product you use
☐	Placed five metrics on the 2×2
☐	Wrote guardrails for two primary metrics
☐	Built the metric architecture assignment
☐	Answered the quiz without notes